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LSTM-Based Online Learning for Real-Time Solar Power Forecasting

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ABSTRACT Accurate solar power forecasting is essential to mitigate the variability and uncertainty of solar power generation and integrate renewable energy sources into power grids. A multi-step Long Short Term Memory (LSTM)-based model is proposed for solar power generation forecasting using online learning. To capture daily fluctuations in solar energy patterns, the model employs a dynamic sunrise–sunset data window operation, ensuring that the relevant transitions in solar irradiance are accurately represented in the training set. For seasonality analysis, the variance of the correlation of features with solar power output is determined across different months of the year. The simulation results show that the proposed LSTM-based online learning model achieves a mean absolute percentage error of less than 1%. The results demonstrate that incorporating solar transition period information significantly enhances the forecasting accuracy of the proposed model. Online learning exhibits higher performance than batch learning in terms of forecasting accuracy.

INDEX TERMS Solar Power, Forecasting, LSTM, Online learning, Dynamic Time windowing.

I. INTRODUCTION

THE global energy demand is growing rapidly, leading to an urgent need to mitigate environmental impacts [1]. This necessity has accelerated the adoption of renewable energy sources such as solar and wind power, which are important for reducing dependence on fossil fuels and minimizing greenhouse gas emissions [2]. However, the integration of these renewable sources into existing power systems poses significant challenges due to their inherent intermittency and unpredictability [3]. Addressing these issues requires highly accurate forecasting models, which are critical for optimizing energy production, maintaining grid stability, and minimizing operational costs [4].

Solar power is one of the most widely utilized renewable energy sources and is integral to achieving global clean energy objectives. Solar power forecasting is essential for effectively integrating solar energy into power grids and optimizing microgrid operations [5]. Solar power generation, primarily influenced by solar irradiance, is subject to variability caused by factors such as temperature, humidity, atmospheric pressure, and cloud cover [6]. These variations directly impact energy production and grid stability, making reliable forecasting indispensable for efficient energy management. Moving beyond these challenges, forecasting accuracy can help minimize reliance on conventional backup systems and reduce operational costs, supporting a sustainable and reliable energy infrastructure.

Long Short-Term Memory (LSTM) networks have been utilized for power prediction due to their ability to learn long-term dependencies and detect complex, nonlinear patterns in time-series data [7]. LSTMs use gating mechanisms to retain only the information relevant to predictions and discard irrelevant data, allowing them to quickly adapt to sudden changes while preserving crucial trends. This makes them effective for short-term solar power forecasting, where rapid fluctuations in output significantly affect grid operations [8]. Different learning approaches, including batch learning and incremental learning, are used to train the LSTM network for real-time solar power forecasting. Batch learning requires retraining the model with accumulated data in batches [9], which is computationally expensive and impractical for real-time applications. Incremental learning, while an improvement over batch learning, updates the model by incorporating new data sequentially but struggles with slow adaptation to sudden changes in solar power generation patterns. In contrast, online learning offers a superior solution by updating

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the model as new data becomes available. Unlike incremental learning, online learning continuously refines the model [10], allowing it to capture real-time variations in solar power generation. This learning approach is particularly beneficial in handling the non-stationarity of solar power data, as it enables rapid adaptation to environmental changes. In addition, online learning mitigates the risk of outdated predictions by ensuring the model remains relevant to the latest data trends.

In this paper, we propose an LSTM-based multi-step prediction model for solar power generation. Online learning is used to continuously update the model's knowledge, maintaining the relevance of its predictions to current solar conditions. The highlights of this work are as follows:

- The proposed model utilizes an LSTM-based online learning framework that continuously updates with new data to forecast real-time solar power generation.
- The proposed model utilizes the time window operation to extract daily solar active periods, leveraging sunrise and sunset information to enhance the model's learning efficiency.
- The relation of different input features, including total solar radiation (TSI), direct normal radiation (DNI), global horizontal radiation (GHI), and air temperature (AT), on solar power generation efficiency is discussed. Furthermore, the impact of input features on the forecasting performance of the proposed model is analyzed through simulations.
- A seasonality analysis is provided by determining the variance of the feature correlation with the solar power output in different months of the year.
- The simulation results demonstrate that the LSTM-based solar power forecasting model has a lower error compared to other machine learning models, including deep learning architectures.
- We also demonstrate that online learning improves forecasting accuracy compared to other training techniques, such as batch learning and incremental learning, due to the continuous updates of the model that enable it to adapt quickly to changes in the solar data.

The remainder of this paper is structured as follows: Section II presents the related works. Section III describes the methodology, including data acquisition, preprocessing, and the development of the LSTM-based online learning model. Section IV presents the evaluation of the proposed methods, including solar power predictions and a comparative analysis of the proposed model with other methods. Finally, Section V concludes the study and discusses potential directions for future research.

II. RELATED WORKS

Traditional forecasting methods, such as statistical models like the Auto-Regressive Integrated Moving Average (ARIMA) [11], have been widely used but come with limitations. These methods often require extensive data and complex computations, making them less practical for real-time applications. Furthermore, their inability to adequately capture the non-linear and random nature of solar irradiance restricts their effectiveness [12]. In response to these limitations, researchers have explored alternative approaches, including physical models based on satellite imagery and meteorological data [13]. Researchers use Support Vector Regression (SVR), an extension of support vector machines, to forecast solar power, incorporating weather variables such as cloudiness to enhance accuracy [14]. However, the SVR method proves useful for specific predictive applications, yet it faces challenges when operating in dynamic forecasting scenarios, as this model does not perform well in detecting extended temporal relationships and changing patterns in large datasets effectively. The characteristics of photovoltaic (PV) power forecasting data, including nonlinearity, volatility, and complex temporal dynamics, make these limitations unsuitable for real-time applications.

The Multilayer Perceptron (MLP) network is used for solar power forecasting because it can handle non-linear relationships [15]. However, it lacks memory mechanisms and sequential data processing capabilities, making it unsuitable for modeling temporal dependencies in time series data. Although Recurrent Neural Networks (RNNs) solve this problem through feedback connections, which maintain information from previous time steps [16], standard RNNs struggle because vanishing gradients create performance issues when processing extended sequences.In [17], the author develops a short-term photovoltaic (PV) power forecasting model through LSTM neural networks, which uses synthetic weather data from past solar irradiance and sky conditions. However, the model shows reduced accuracy during unstable weather conditions, particularly when cloud cover undergoes sudden changes.

Recent research has focused on advanced deep learning architectures, including Convolutional Neural Network Long Short-Term Memory (CNN-LSTM) and Bidirectional Long Short-Term Memory (BiLSTM) [18]. The CNN-LSTM model uses convolutional layers to detect local patterns before learning temporal features. The convolutional layers can over-extract local features, leading to redundant feature extraction. The BiLSTM model utilizes bidirectional sequence processing to enhance representation; however, it introduces noncausal dependencies that are unsuitable for real-time solar forecasting. In addition, the researcher also explored metaheuristic optimization techniques [19], such as the reptile search algorithm and modified particle swarm optimization [20] for hyperparameter tuning of LSTM networks. These optimization techniques face limitations due to their high computational requirements, limited population sizes, and iteration constraints. Constructive and destructive approaches are also utilized to tune network hyperparameters [21], [22], as they are resource-efficient and facilitate greater control and interpretability of the model.

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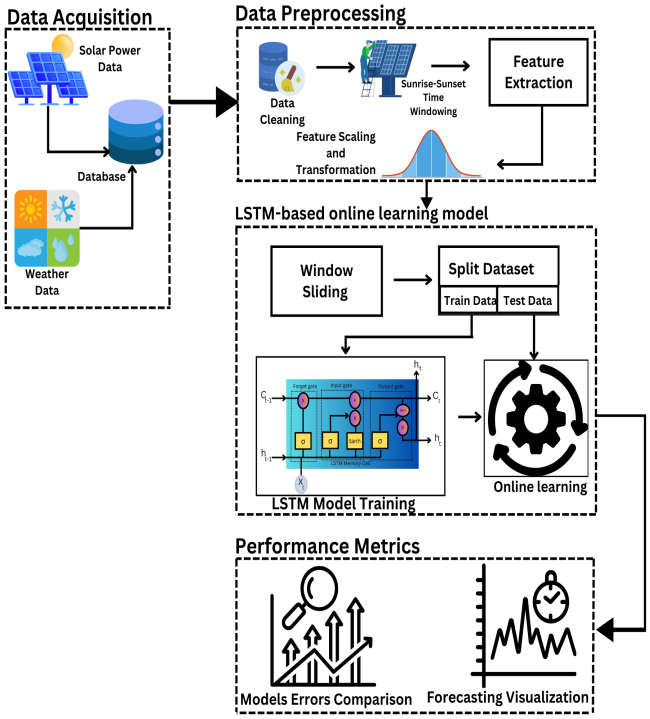


FIGURE 1. **Methodology for solar power forecasting involves data acquisition, data preprocessing, LSTM-based online learning model, and performance evaluation through forecast visualization and error comparison.**

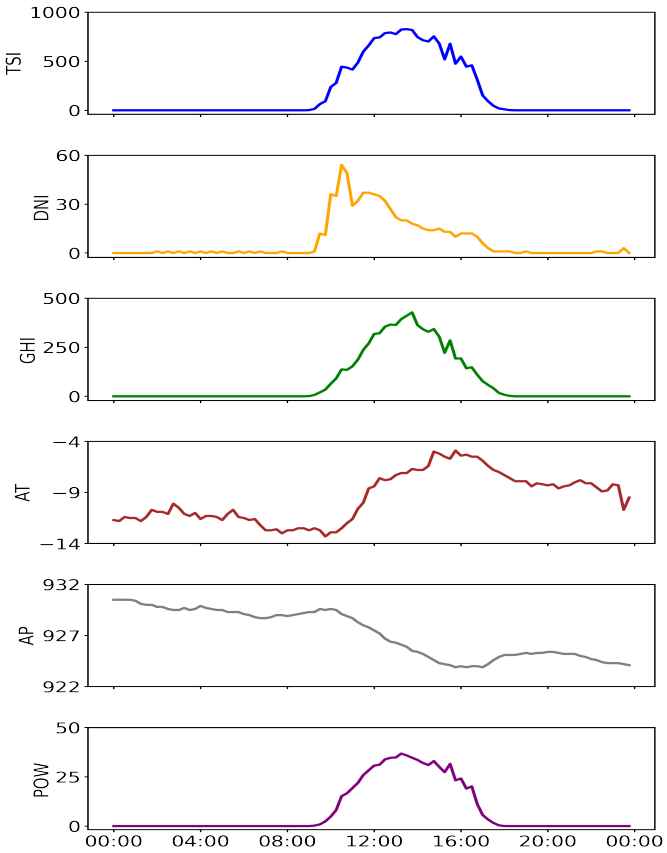


FIGURE 2. **The visualization shows the trends of key solar power and environmental parameters of a single day. Each subplot illustrates how various variables influence the behavior of solar power generation (output) throughout the day.**

III. METHODOLOGY FOR FORECASTING SOLAR POWER GENERATION USING LSTM

An LSTM-based online learning forecasting method is proposed for solar power generation. The proposed methodology, as shown in Figure 1, involves the acquisition of solar data, its preprocessing, the design of an LSTM network, and the implementation of an online learning mechanism. Data from solar power plants is collected and preprocessed for feature extraction. The LSTM neural network, designed and trained using the extracted features through online learning, forecasts the power generation of solar plants. Each component of the proposed methodology is detailed in the following subsections.

A. DATA ACQUISITION

The solar data collected from solar power plants containing on-site measurements [23], includes solar features TSI, DNI, and GHI, and environmental factors such as AT and AP. The solar stations were located in various regions of China, and the measured temperature was between 18.2°C and 41.2°C. Figure 2 illustrates the daily variations in solar and environmental parameters, where each parameter exhibits a distinct pattern. They peak around noon and are directly correlated with solar power. AT steadily increases from morning to afternoon before dropping later in the day, following a typical daily cycle. AP gradually decreases throughout the day, likely influenced by temperature fluctuations. Overall, power output closely mirrors solar irradiance, increasing after sunrise, peaking at midday, and tapering to zero at sunset. This emphasizes the direct relationship between solar irradiance and energy generation.

B. DATA PREPROCESSING AND FEATURE SELECTION

Data preprocessing involves extracting solar transition periods using sunrise and sunset times, as well as features that influence solar power output. The sunrise and sunset information is utilized to define a time window for the day ω. The window ω represents a specific period of the day, which is defined as

$$T_r - \theta \leq \omega \leq T_s + \theta,$$

where T_r and T_s are the sunrise and sunset times, respectively. θ is a positive integer value added to sunrise and sunset times to facilitate the inclusion of critical solar transition periods. The window is applied to the acquired data, allowing the data to be collected over a period of ω for each day, thus adapting to variations in solar daily trends. Figure 3 shows an example of the sunrise-sunset time window. Due to the different seasons in January and July, the duration of sunlight varies on the days of these months, and the length of ω is different on both days. The window operation provides the model with valuable

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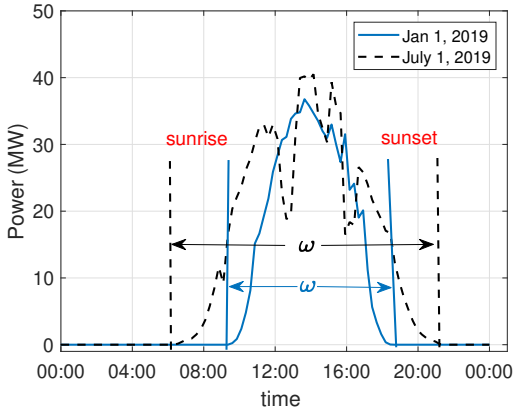


FIGURE 3. The visualizations show power generation on January 1 and July 1, 2019.

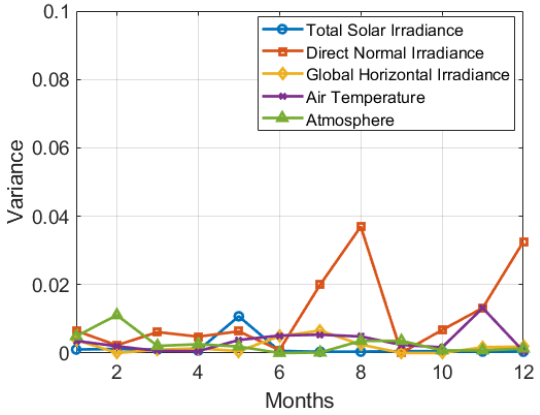


FIGURE 4. Variance of feature correlations with power output across 12 months.

TABLE 1. Correlation of features with power output

	TSI	DSI	GHI	AT	AP
POW	94.6%	57.9%	64.9%	22.8%	-5.7%

information on how to learn and adapt to variations in solar energy trends.

The feature selection aimed to investigate the key factors, including TSI, DNI, GHI, AT, and AP, that influence solar power generation. To assess how closely these features are related to the solar power output, we used the correlation coefficient $\rho_{x,y}$ between two variables x and y [24]. The formula for calculating this coefficient is:

$$\rho_{x,y} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \cdot \sum_{i=1}^{n}(y_i - \bar{y})^2}},$$

where x_i and y_i are the observed values, and $\bar{x}$ and $\bar{y}$ are the respective means of x and y. The correlation between feature data and the solar power output is shown in Table 1, which provides information on the relevant feature for the solar power generation forecasting model. Table 1 shows that four of the five parameters are the most influential factors affecting solar power generation, which is also observed in Figure 2. TSI, DNI, and GHI represent the energy available for solar panels, while AT impacts power generation efficiency. AP does not influence solar power generation because its correlation value is negative.

For seasonality analysis, we determine the variance of correlation across different months of the year to see the influence of different seasons (i.e., temperature) on the relationships between features and solar power output. The variance of correlation between a feature and the power output for the j-th month of the year, ϱ^i, is calculated as [25]:

$$\varrho^j = \mu - \rho^j,$$

where μ represents the mean of the correlation values of 12 months and ρ^j indicates the correlation value for the j-th month. Figure 4 shows the variance of the correlation values for each feature over a year. None of the selected features exhibits higher variance in any of the 12 months. The highest correlation variance is 0.037 for DNI in August, a relatively small value that indicates that the influence of the selected features on power output remains fairly consistent. This ensures that the forecasting model can effectively learn patterns without needing adjustments for different seasons.

Normalization is applied to mitigate disparities in value ranges, ensuring equitable contributions from selected features. This prevents features with larger values from dominating the model's learning process. Min-max normalization is used to standardize all feature values within a consistent range [21], enhancing the model's ability to interpret and learn from the data effectively, as follows:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}},$$

where X_{norm} is the normalized value of the feature, X is the original value of the feature, $X_{\min}$ is the minimum value of the feature, and $X_{\max}$ is the maximum value of the feature in the dataset.

C. PROPOSED LSTM TRAINING MECHANISM

LSTM is designed and trained to forecast solar power generation by capturing long-term dependencies within sequential data. LSTM, a type of RNN, addresses the problem of vanishing gradients and allows the retention of information over long sequences [26]. The architecture of an LSTM, shown in Figure 5, consists of three gates: forget, input, and output, which regulate the flow of information and effectively manage the cell state. The forget gate determines which information from the previous cell state should be discarded, while the input gate identifies new information to be added to the cell state. Together, these gates update the cell state, enabling the model to retain relevant information and discard unnecessary details. The output gate then generates the hidden state, which is derived from the updated cell state and serves as the final

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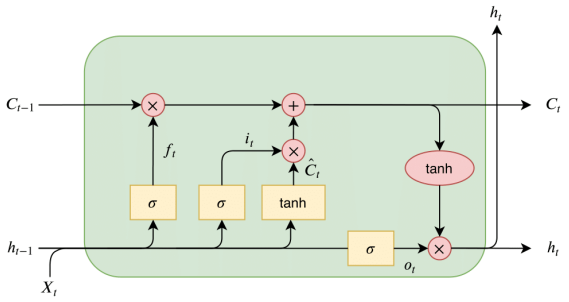


FIGURE 5. LSTM architecture: X_t is the input time step, h_t is hidden state, C_t is the cell state, f_t is the forget gate, i_t is the input gate, o_t is the output gate, and $\hat{C}_t$ represents the candidate cell state.

output of the cell. The computations for the LSTM gates and states are as follows,

$$
\begin{aligned}
f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \\
i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \\
\tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C), \\
C_t &= f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t, \\
o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \\
h_t &= o_t \cdot \tanh(C_t),
\end{aligned}
$$

In these equations, f_t, i_t, and o_t represent the output of the forget gate, the input gate, and the output gate, respectively. C_t is the updated cell state, h_t is the hidden state, and $\tilde{C}_t$ is the candidate cell state. W and b denote weights and biases, respectively. σ and $\tanh$ represent the sigmoid and hyperbolic tangent activation functions, respectively.

The proposed method trains the LSTM model on an initial training set, and then *online learning* is used to continuously update the model's knowledge as new data become available, ensuring that its predictions remain relevant to current conditions. Once the initial training is complete, the LSTM model forecasts two hours ahead of solar power generation, and retraining occurs on a new data sequence. The model is updated by retraining on the new sequence following online learning. This allows the LSTM model to effectively adapt to fluctuations in solar power generation. The proposed method will achieve reliable real-time solar power forecasting by continuously incorporating recent data and fine-tuning the model parameters.

IV. EVALUATION OF PROPOSED METHOD

A. EXPERIMENTAL SETUP

The dataset [1] for this research was collected from a 50 Megawatt (MW) solar power plant, which records information on solar power and weather conditions at 15-minute intervals. The simulations were performed using five months of solar power data. The first four months of data, from November 2019 to February 2020, were used for the initial training of the LSTM model, while the remaining data from March 2020 were used for online learning. A sliding window technique [27], illustrated in Figure 6, is employed to train the LSTM network, which transforms time series data into input sequences and discards outdated patterns. The sliding window has a size of 128 with a recurrent step size of one. The LSTM architecture comprises three LSTM layers with 100, 85, and 75 units, using a descending configuration to decrease the complexity of the model while preserving its ability to learn temporally. The learning rate was set to 1e-3, which was determined to find the optimal balance between convergence speed and stability. The initial training process used a batch size of 16 to provide efficient and stable gradient updates. The model performs real-time updates with each new data point that enters the system, since it uses online learning to maintain continuous adaptation to recent trends. The model runs 300 epochs of initial training before receiving online updates through 50-epoch cycles for each new sequence. The hyperparameters for the LSTM network are summarized in Table 2.

[1]The dataset was used in the Renewable Energy Generation Forecasting Competition hosted by the Chinese State Grid in 2021 [23].

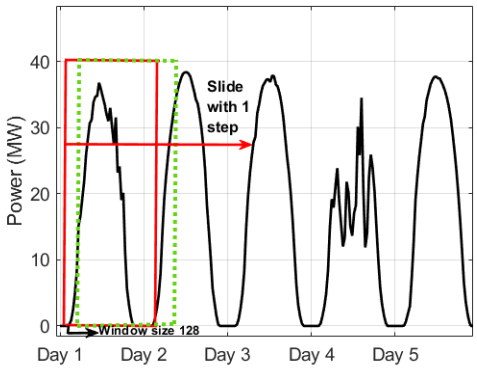


FIGURE 6. Window sliding operation.

TABLE 2. Hyperparameter and training configurations.

Parameter	Value
Input features	4
Hidden dimensions (Layer 1, Layer 2, Layer 3)	100, 85, 75
Learning rate (initial, online)	1e-3
Batch size	16
Number of epochs (initial, online)	300, 50
Optimizer	Adam
Loss function	MSE
θ	2 hours

B. PERFORMANCE EVALUATION METRIC

Several performance metrics, including normalized error, MAPE, absolute error, Root Mean Square Error (RMSE), and the coefficient of determination (R^2), were used to assess the accuracy of the proposed LSTM-based forecasting model.

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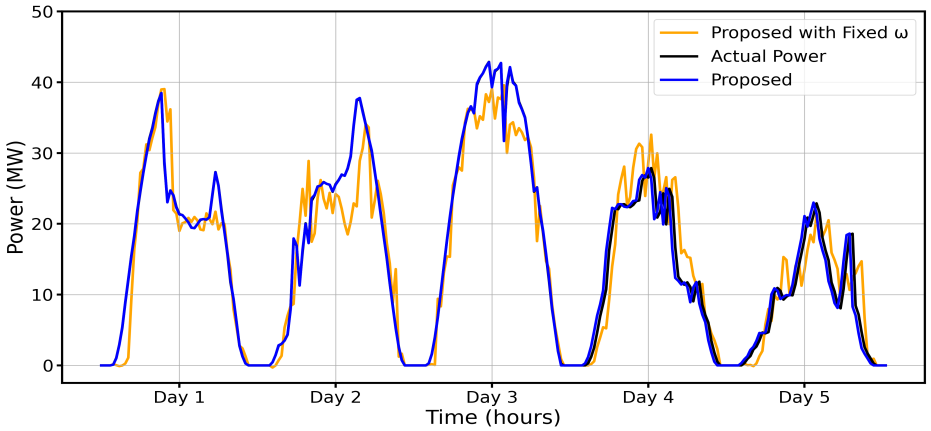


FIGURE 7. Actual power and predicted power of the proposed model.

Normalized error is calculated for each individual day as

$$\text{Normalized Error} = \left(\frac{P_{\text{act}} - P_{\text{pred}}}{P_{\text{act}}}\right) \times 100,$$

where P_{act} and P_{pred} represent the actual and predicted solar power values. Normalized error provides the daily error as a percentage relative to the actual power, allowing for day-to-day comparison of the accuracy of the model. MAPE is calculated as the accumulated error over multiple days by averaging the daily normalized errors over n days.

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{P_{\text{act},i} - P_{\text{pred},i}}{P_{\text{act},i}}\right| \times 100,$$

Absolute errors are calculated as

$$\text{Absolute Error} = |P_{\text{act}} - P_{\text{pred}}|.$$

The lower error distribution reflects a close match between the actual and predicted values, which underscores the accuracy of the model in forecasting solar power. RMSE measures the standard deviation of the prediction errors and is given as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2},$$

where y_i denotes the total actual power, $\hat{y}_i$ is the total predicted power, and n represents the number of days. while the value of R^2 represents the percentage of the dependent variable used by the model. It is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2},$$

where $\bar{y}$ is the mean of the total actual power. The value of R^2 ranges from 0 to 1, with values closer to 1 indicating a stronger model fit.

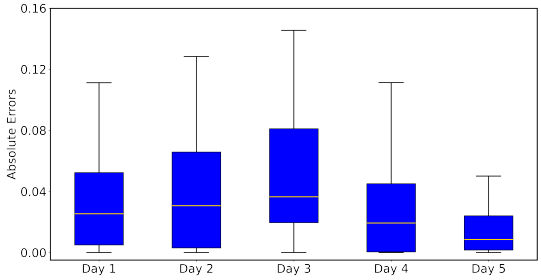


FIGURE 8. Prediction error of the proposed model.

C. FORECASTING ACCURACY OF THE PROPOSED METHODOLOGY

The performance of the proposed model in predicting solar power is shown in Figure 7. The proposed model demonstrates high accuracy in solar power forecasting, accurately predicting power levels close to the actual solar power. The proposed model continuously adjusts its predictions using online learning, effectively capturing variations in solar power generation. The proposed model with fixed ω shows significant deviations from actual power during peak power intervals. The unnecessary zero values during sunrise and sunset periods lead to increased errors. The proposed model, which incorporates a dynamic time window, eliminates irrelevant zero values, resulting in significantly improved forecast accuracy. A boxplot (showing the interquartile ranges) of absolute errors under the proposed model with a dynamic time window ω is presented in Figure 8. A lower distribution of errors indicates a closer match between the actual and predicted values, emphasizing the accuracy of the proposed model in forecasting solar power. The small interquartile range observed under the proposed model shows that errors remain consistently low, indicating the stability of the model.

To demonstrate the generalization of the proposed trained model, a scatter plot is shown in Figure 9, which compares actual power values with the predicted power values in the

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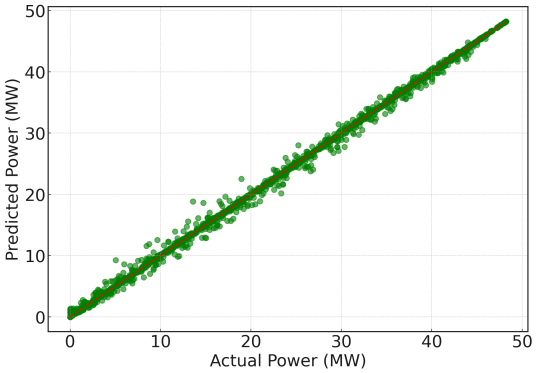


FIGURE 9. Comparison of LSTM-based online learning prediction with real data for test set.

TABLE 3. Average absolute error for the proposed model.

Month	Average Absolute Error
March	0.269
June	0.279

test set. The data points in Figure 9 form a tight cluster around the diagonal line, indicating a smaller difference between the actual and predicted values. This indicates that the proposed model exhibits a higher and consistent predictive accuracy throughout the test set, demonstrating its strong generalizability.

To assess the impact of seasonal variation on the proposed model, we evaluated the performance of the trained LSTM with online learning in the months of March and June, specifically during the spring and summer seasons. The average absolute error for both months is shown in Table 3. The proposed forecasting model has very small and approximately the same errors in both months, indicating that it continuously learns seasonal patterns and achieves higher prediction accuracy in varying seasons. In conclusion, the proposed approach ensures accurate and reliable solar power predictions by adapting to solar patterns through continuous learning.

D. IMPACT OF DIFFERENT FEATURES ON THE PROPOSED MODEL

The impact of different features, discussed in subsection III-B, on the forecasting performance of the proposed model is investigated by training the LSTM network with various feature sets. Figure 10 shows the error of the trained LSTM network with different feature sets over five days of solar power forecasting. When the LSTM has four input features, TSI, DNI, GHI, and AT, it shows the lowest normalized error for three days compared to when the LSTM uses fewer features, seen in Figure IV-D. The LSTM with three input features (TSI, DNI, and GHI) shows a similar performance to that when the LSTM uses all four features as input, achieving slightly lower error on the third and fourth days. However, on day five, the error of the LSTM with three input features increases from 0.025 to 0.366, while the LSTM with all four

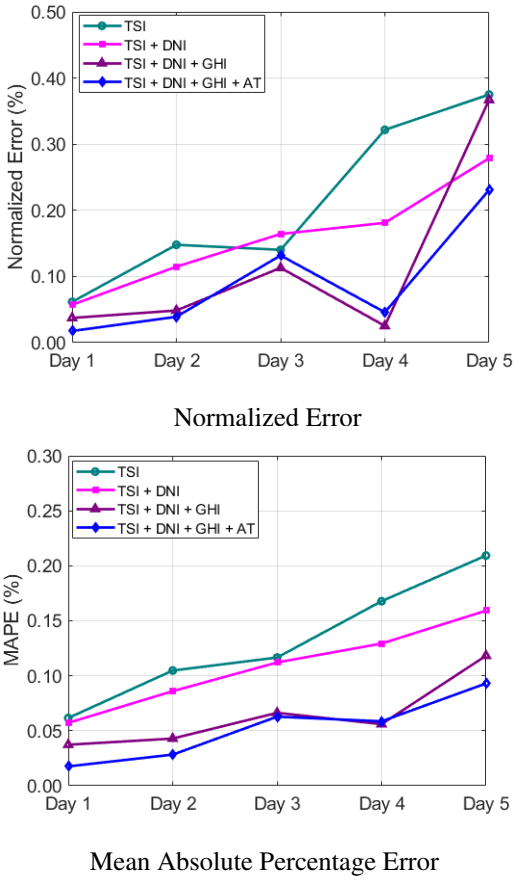


FIGURE 10. Comparison of different feature sets on solar power forecasting accuracy.

input features has a normalized error equal to 0.230.

The LSTM with the only TSI feature has higher errors on most days than the LSTM with more input features, indicating that incorporating additional parameters can improve predictions. Similar performance trends are observed in Figure IV-D, which shows the MAPE for the LSTM network with different input features. Using AT along with the other three input features (TSI, DNI, and GHI) slightly improves the forecasting performance of the LSTM network compared to when AT was not used as an input feature. Although TSI has the highest correlation with output power, as shown in Table 1, using TSI as the sole input feature yields the highest errors throughout. Using all four parameters as input features for solar power forecasting enhances accuracy and stabilizes forecast performance, especially over longer prediction horizons.

E. IMPACT OF TRAINING TECHNIQUES ON THE PROPOSED MODEL

The performance of the proposed forecasting model under various training techniques, including online learning, incremental learning, and batch learning, is evaluated for solar power forecasting and is shown in Figure 11. The proposed model with online learning provides more accurate forecast-

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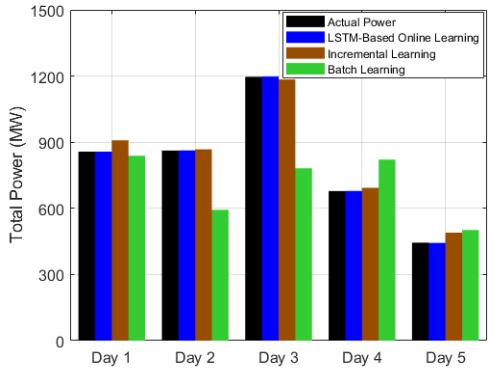


FIGURE 11. Performance of the proposed model with incremental learning and batch learning.

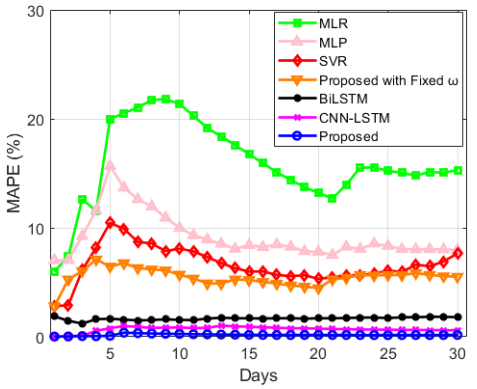


FIGURE 12. MAPE comparison of different models with the proposed model over 30-day prediction.

TABLE 4. Time complexity comparison between online and batch learning on the test set.

	Online learning	Batch learning
Total Time	14.6 mins	758.4 mins

ing of solar power generation, whereas batch learning exhibits a higher mismatch between actual and estimated values. Batch learning, a static learning method that is retrained daily, causes the proposed model to fail in incorporating new trends from solar data, resulting in large forecasting errors. Although the incremental learning model (with a batch size of 16 retrained three times a day) demonstrates improved accuracy through periodic updates with new data daily, it shows delays in adapting to daily fluctuations, resulting in moderate prediction performance. The proposed LSTM-based online learning continuously updates, thus accurately capturing ongoing variations in solar irradiance and yielding total power predictions close to the actual output.

For the time complexity comparison between online learning and batch learning, the total time required to forecast 30 days of test data following initial training, as presented in Table 4. The model employing online learning required only 14.6 minutes to sequentially predict and retrain on the test dataset. In contrast, the batch learning strategy was approximately 52 times slower. In the batch learning setup, after each day's forecast is completed, the corresponding test data is appended to the training dataset. The model is then retrained from scratch on the accumulated data before proceeding to forecast the next day. This repetitive full retraining process significantly increases time complexity as the dataset grows. The online learning updates the model with the most recent data, avoiding full retraining and making it suitable for real-time forecasting applications with continuous data streams.

F. COMPARISON WITH OTHER MACHINE LEARNING MODELS

The effectiveness of the proposed LSTM-based online learning model is validated by comparing its performance with machine learning algorithms, including MLP [28], SVR [29], and multilinear regressor (MLR) [30], as well as deep learning models based on LSTM such as CNN-LSTM and BiLSTM [18].

The MAPE values of different forecasting models are shown in Figure 12. The proposed model demonstrates the lowest MAPE in all 30 days by maintaining it below 1%, indicating its superior accuracy and stability. Traditional methods, SVR, MLP, and MLR, lack the sequential modeling ability necessary to capture temporal dependencies and, therefore, result in significantly higher errors. The proposed model consistently achieves lower errors compared to CNN-LSTM and BiLSTM because of its simple architecture and strong temporal modeling capabilities. The MAPE value for the proposed model is less than 0.2, whereas for CNN-LSTM, the second-best forecasting model, it is higher than 0.5. For BiLSTM, the MAPE value ranges between 1.5 and 1.8, which is higher than the proposed model (LSTM) and CNN-LSTM but much lower than the non-LSTM-based forecasting models. The proposed method with fixed ω performs poorly because it includes zero data (e.g., nighttime values), which introduces noise. This reduces the model's ability to learn meaningful solar patterns.

Table 5 shows the normalized error, MAPE, RMSE, and R^2 average of the 30 days of different forecast models. The proposed LSTM-based online learning model achieves the highest performance in all evaluation metrics, delivering the lowest normalized error of 0.16, the lowest MAPE of 0.17, the lowest RMSE of 4.72, and the highest R^2 of 0.99958, on average, over 30 days of test data. The proposed model employed a unidirectional LSTM architecture to efficiently detect long-term temporal dependencies while maintaining low complexity and achieving rapid convergence.

V. CONCLUSION

This paper proposed an LSTM-based online learning model for multi-step solar power forecasting. The model utilizes a dynamic sunrise–sunset window to effectively capture daily variations in solar periods. A seasonality analysis is provided

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TABLE 5. Performance comparison of different forecasting models.

Model	Normalized Error	MAPE	RMSE	R^2
Proposed	**0.16**	**0.17**	**4.72**	**0.99958**
CNN-LSTM	0.57	0.72	8.21	0.99830
BiLSTM	1.75	1.71	13.34	0.99480
Proposed with Fixed ω	5.17	5.53	69.94	0.91494
SVR	7.26	6.53	93.82	0.84980
MLP	7.35	8.80	89.39	0.86360
MLR	17.21	15.22	179.86	0.44800

by determining the variance of the feature correlation with the solar power output, which gives a very small correlation variance for all months of the year. The proposed model showed superior performance compared to CNN-LSTM, BiLSTM, and traditional machine learning models, with an MAPE of less than 1%. The results showed that the proposed forecasting model has very small prediction errors for the spring and summer months, indicating its robustness to seasonal variations.

The performance of the proposed model is prone to faulty sensor readings, including missing information or noise. Future work will focus on designing a forecasting system that detects faulty data and prevents learning from such data, thereby providing a reliable estimate of renewable energy generation.

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